

OSLAB SKILL-3 PROGRAMS & OUTPUT

Date: 09-09-2026

Program-1 Code & Output:

```
sainath@Sainath:~/OSLAB/skill3$ gedit program1.c
sainath@Sainath:~/OSLAB/skill3$ gcc program1.c -o program1
sainath@Sainath:~/OSLAB/skill3$ cat program1.c
#include <stdio.h>
#include <string.h>

int main() {
    char history[5][50];
    int n = 0, choice;

    printf("Enter 5 commands:\n");

    while (n < 5) {
        printf("> ");
        fgets(history[n], 50, stdin);
        history[n][strcspn(history[n], "\n")] = '\0';

        if (strcmp(history[n], "exit") == 0)
            break;

        n++;
    }

    printf("\nCommand History:\n");
    for (int i = n - 1; i >= 0; i--)
        printf("%d: %s\n", i + 1, history[i]);

    return 0;
}
sainath@Sainath:~/OSLAB/skill3$ ./program1
Enter 5 commands:
> ls
> pwd
> date
> whoami
> gcc

Command History:
5: gcc
4: whoami
3: date
2: pwd
1: ls
```

Program-2 Code & Output:

```
sainath@Sainath:~/OSLAB/skill3$ gedit program2.c
sainath@Sainath:~/OSLAB/skill3$ gcc program2.c -o program2
sainath@Sainath:~/OSLAB/skill3$ cat program2.c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, i, *a;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    a = malloc(n * sizeof(int));

    if (a == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Elements: ");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);

    free(a);

    printf("\nMemory released successfully\n");

    return 0;
}
sainath@Sainath:~/OSLAB/skill3$ ./program2
Enter number of elements: 4
Enter 4 elements:
10 20 30 40
Elements: 10 20 30 40
Memory released successfully
```

History:

```
384 cd OSLAB
385 ls
386 mkdir skill3
387 cd skill3
388 gedit program1.c
389 gcc program1.c -o program1
390 ./program1
391 gedit program1.c
392 gcc program1.c -o program1
393 cat program1.c
394 ./program1
395 gedit program2.c
396 gcc program2.c -o program2
397 cat program2.c
398 ./program2
399 history
```